

SELECTED WORK · 2024-2026

# Charles Abi Chahine

architect · design technologist

computational design · BIM · AI in architecture

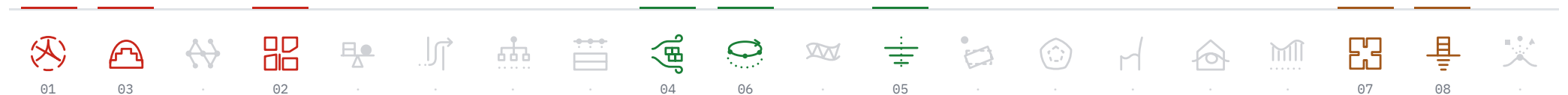
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Architect and design technologist, Master in Advanced Computation for Architecture and Design (MaCAD) at IAAC. My focus is computational design in service of architecture: parametric design, BIM delivery and AI-assisted workflows, with a particular interest in digital fabrication.

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## Sensi

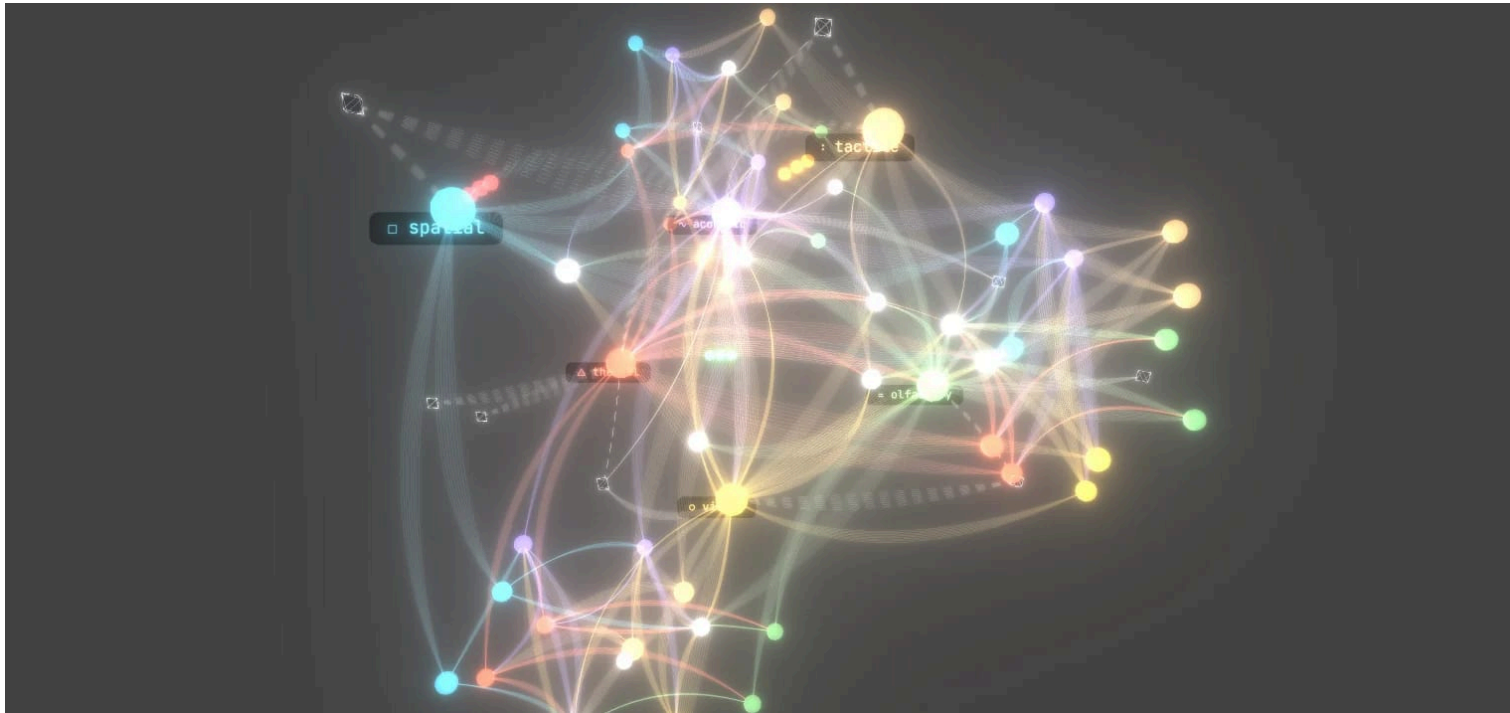
2026

COMPUTATION & AI

MACAD 2026 AWARD

### A sensory copilot for floor plans.

In architecture we model everything: structure, cost, energy, code compliance. Layer after layer that makes a building accountable before it is built. Sensi adds the one layer we never formalized: how a space will actually feel. Not emotion, but the full sensory experience of standing in a room, scored across six senses: thermal, visual, acoustic, spatial, olfactory, tactile. Not in the abstract, but for a specific person.



#### MODULE

AIA Studio · MaCAD, IAAC

#### TEAM

Charles Abi Chahine, Emilie El Chidiac,  
Lakzhmy Mari Zaro, María Sánchez  
Dominguez

#### TOOLS

Python · LLM agents · Vision models ·  
Web app

#### LIVE

sensi.emiliechidiac.com

## 01 · THE SENSORY LAYER

Comfort research studies one sense at a time, but we take a room in through all of them at once, and they are coupled: one moderates another. The room score is half mean, half worst, because the worst sense is the one you would actually feel. Scoring is deterministic; the LLM explains results, it never decides them.

02 · ACT 1 · ONBOARD: WHO IS THIS FOR

Comfort is not universal, so Sensi opens by learning who you are: a few questions, then a moodboard where each image you keep quietly tags the senses you lean toward. It compiles into a persona drawn as a petal rose.

## 03 • ACT 2 • SHAPE: COMFORT YOU CAN EDIT

The heart of the system. You talk to it in plain language and a fast routing model classifies your intent in about 0.6 seconds. An edit is planned, validated, applied and re-scored live, kept as a checkpoint. Rooms are nodes and doors are edges, so comfort becomes a zoning problem you can see.

## 04 • ACT 3 • REPORT: CLOSING THE LOOP

Final scores write a prompt and a vision model renders each room under an honest rule: only the extreme senses speak, the comfortable middle stays quiet. About \$0.039 per room, 2.75× faster than the alternative we tested.

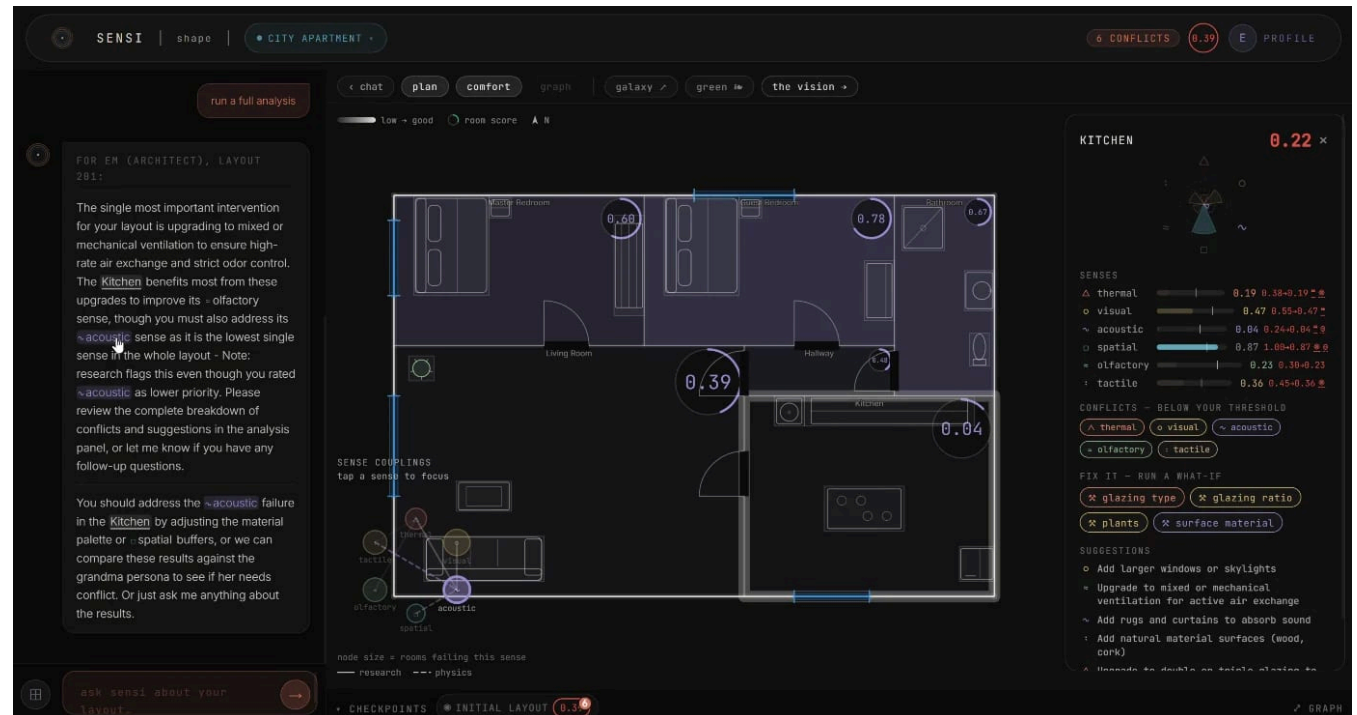


FIG 01.1

A full analysis: the kitchen flagged, the fixes suggested, the whole layout read.

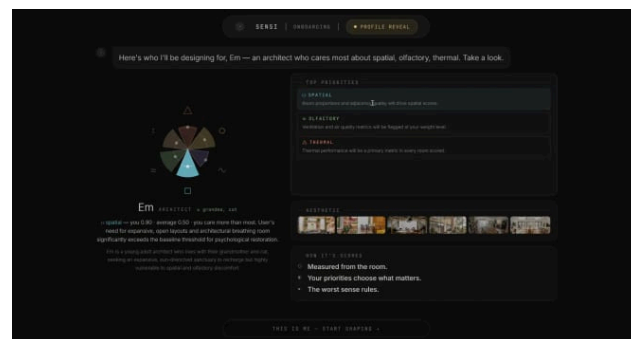


FIG 01.2

The profile reveal: who Sensi is designing for, and how the plan will be scored.

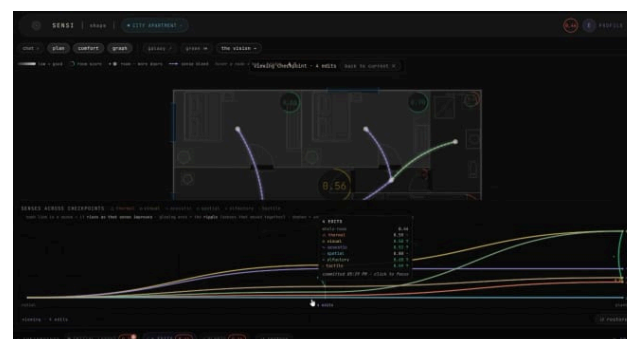


FIG 01.3

Senses across checkpoints: every edit committed, every sense a line that answers it.

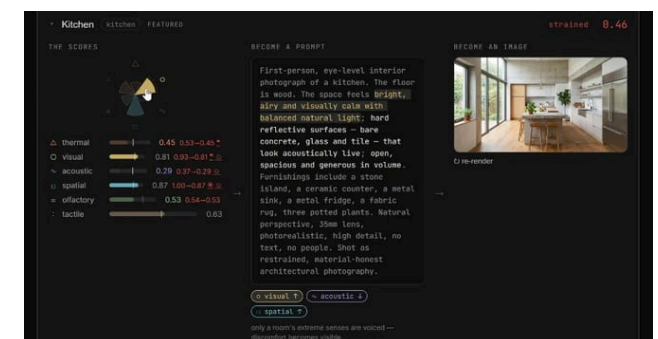


FIG 01.4

Room-by-room detail, ready for a design team to act on.



# Encoding Urban Risk

2026

COMPUTATION & AI

Street risk classified from open map data.

Can the physical layout of a street, measurable from public map data, predict how risky it feels? We framed it as a three-class problem: low, medium, or high risk, using seven spatial features drawn from OpenStreetMap and Mapillary per street segment. This is the full arc of the project, not just the results, but the wrong turn that made them honest.



1961 · Jacobs

**ALIVE TO BE SAFE**

Eyes on the street, active frontages, mixed uses & foot traffic create natural surveillance.



1972 · Newman

**OWNED TO BE SAFE**

Territorial control, restricted access. Small recognisable communities self-police.



1984 · Hillier & Hanson

**LAYOUT DETERMINES LIFE**

Street configuration shapes movement, encounter and territorial control — Space Syntax.



1984 · Brantingham

**FAMILIARITY BREEDS CRIME**

Offenders strike within their awareness space — familiar routes and nodes.



2008 · Hillier & Sahbaz

**MORE NEIGHBOURS = SAFER**

Residential density neutralises risk. Safety in numbers confirmed across 65,000+ dwellings.



2017 · Summers & Johnson

**ARTERIES & SHADOWS**

Global Choice is the strongest single predictor. Crime spills within 100 m of major routes.

## MODULE

Data Encoding · MaCAD, IAAC

## TEAM

Charles Abi Chahine, Emilie El Chidiac, Lakzhmy Mari Zaro, María Sánchez Domínguez

## TOOLS

Python · scikit-learn · OpenStreetMap · Mapillary · SHAP

## 01 · GROUNDED IN SIXTY YEARS OF THEORY

Every feature traces to established urban-safety theory: Jacobs on eyes on the street, Newman on territorial clarity, Hillier and Hanson on network configuration. The unit is the segment, where risk clusters.

## 02 · HITTING THE WALL

We first tried to predict crime directly, regressing 920,000 London incidents against our features. Three models returned  $R^2$  under 0.064: street geometry cannot capture social forces. A finding, not a failure.

## 03 · THE PIVOT: A SPATIAL TYPOLOGY

Rebuilt in eight steps: fetch from OSM and Mapillary, seven normalised features per segment, collapsed into a weighted score, cut into three classes with K-Means, then trained. 31% low, 47% medium, 22% high.

## 04 · COHERENT TYPES, AMBIGUOUS RISK

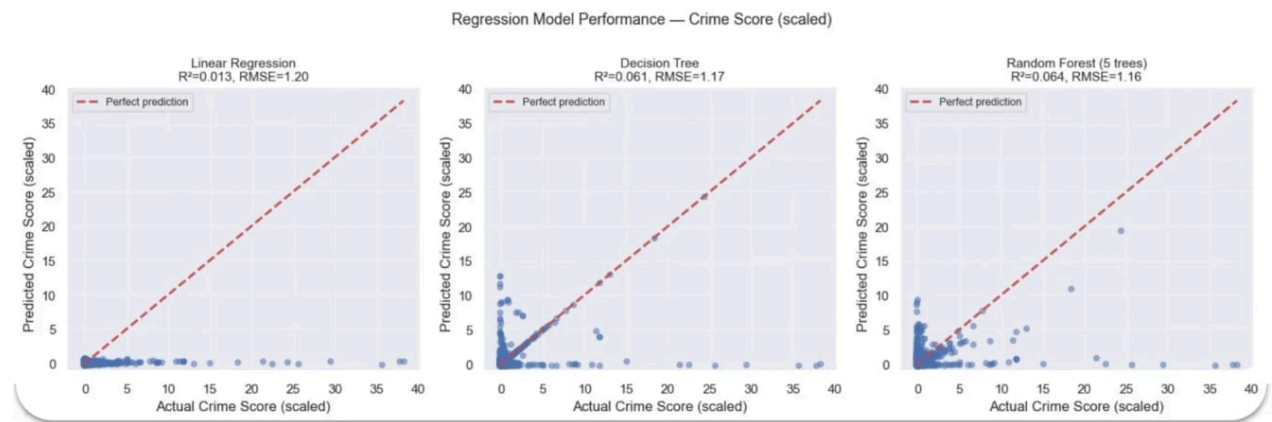
The Kohonen map organises into legible street types, but coloured by risk, near-identical cells differ sharply. Typologies describe character; they do not determine risk. That gap is the honest heart of it.

## 05 · WHAT ACTUALLY MATTERS

Logistic Regression hits 99% and Random Forest 95%, expected since the label came from the same features. The ablation says more: transport leads, then connectivity and land use. Lighting barely registers.

## 06 · LONDON TRAINED, WORLD TESTED

Applied to Barcelona's Eixample and Trastevere, the reading breaks where morphology diverges. SHAP pinpoints it: land use and visibility hit values never seen. A distribution problem, not an architecture flaw.



Crime is too **complex** a phenomenon to **reduce** to spatial features only.  
What we can interpret is **risk** based on & stemming from **geometrical phenomena**.

FIG 02.1

The wall: every model flat against the diagonal. Crime does not reduce to spatial features.



FIG 02.2

— low — medium — high risk One model, three fabrics: the reading breaks where the morphology diverges.



# LEgoarCh

2026

COMPUTATION & AI

IAAC EXHIBITION

A prompt becomes a buildable  
LEGO set.

You type a building, and a minute later a real LEGO set is sitting on a shelf: rendered, modelled, brick-built, priced, and catalog-legal. Most "AI makes LEGO" demos stop at a gorgeous render, but a render is a promise, not a product. LEgoarCh builds the downstream half: the machinery that forces the dream to obey real bricks, real colours, and real gravity.



## AI PROPOSES PROBABILISTIC

01



**Prompt**

a building name

02



**FLUX.2 + LoRA**

render · 28 steps

03



**TRELLIS-2**

image to 3D mesh

## COMPUTATION PROVES DETERMINISTIC

04



**Voxelize**

32 studs wide

05



**Legolize**

split + merge

06



**Buildable set**

stable & legal

## MODULE

Generative AI Seminar · MacAD, IAAC

## TEAM

Charles Abi Chahine, Emilie El Chidiac

## TOOLS

FLUX.2 + LoRA · TRELLIS-2 · ComfyUI · Python

## LIVE

legoarch.charlesabichahine.com

**01 · ONE SENTENCE, FIVE MOVES**

Text prompt, FLUX render, TRELIS mesh, voxelize, legolize, a priced buildable set. The front half is probabilistic and invents the form. The back half is deterministic and proves it stands. The handoff in the middle is the entire idea.

**02 · A LORA THAT SPEAKS LEGO**

Base FLUX.2 does not speak fluent LEGO Architecture. We trained a LoRA on the visual grammar of real LEGO Architecture sets and swept its strength: at 1.0 the studs and brick seams snap in. The LEGO-ness lives in the fine-tune, not in the words.

**03 · FROM ONE PHOTO TO A WHOLE OBJECT**

TRELIS-2 completes the full 3D form from the single render, openly guessing the unseen back. Then the deterministic half begins: the mesh is voxelized onto plate-height layers, pre-stretched 2.5× vertically because bricks are not cubes.

**04 · THE LEGOLIZER: WHERE BUILDABLE GETS EARNED**

The computational centrepiece contains no AI at all. A split-and-merge solver covers each layer with the largest legal bricks that fit, with a slope pass and a running-bond penalty. 44 parts by 48 colours give 1,598 validated combinations.

**05 · A SET YOU CAN KEEP**

The output is a product: box art, a build booklet, a priced parts list, and a shelf that keeps every set.



FIG 03.1

Same prompt, same seed, only the LoRA strength changes: at 1.0 the studs and seams snap in.

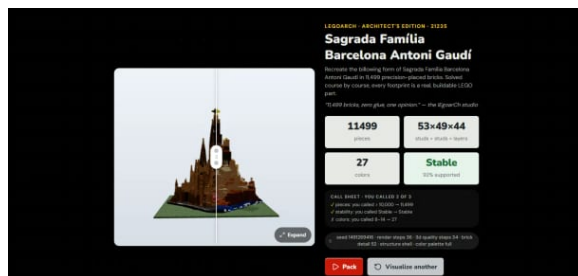


FIG 03.2

The builder, mid-handoff: the smooth mesh on one side of the slider, the bricks on the other.

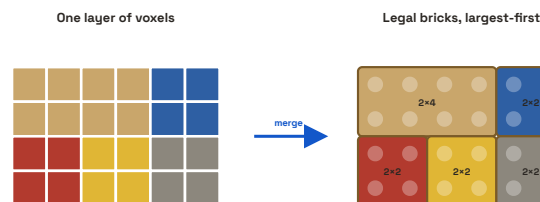


FIG 03.3

Split and merge, one layer: the biggest real bricks that share a colour, not a sea of 1x1s.



FIG 03.4

Box art and build booklet, generated per set.



# The Huddle

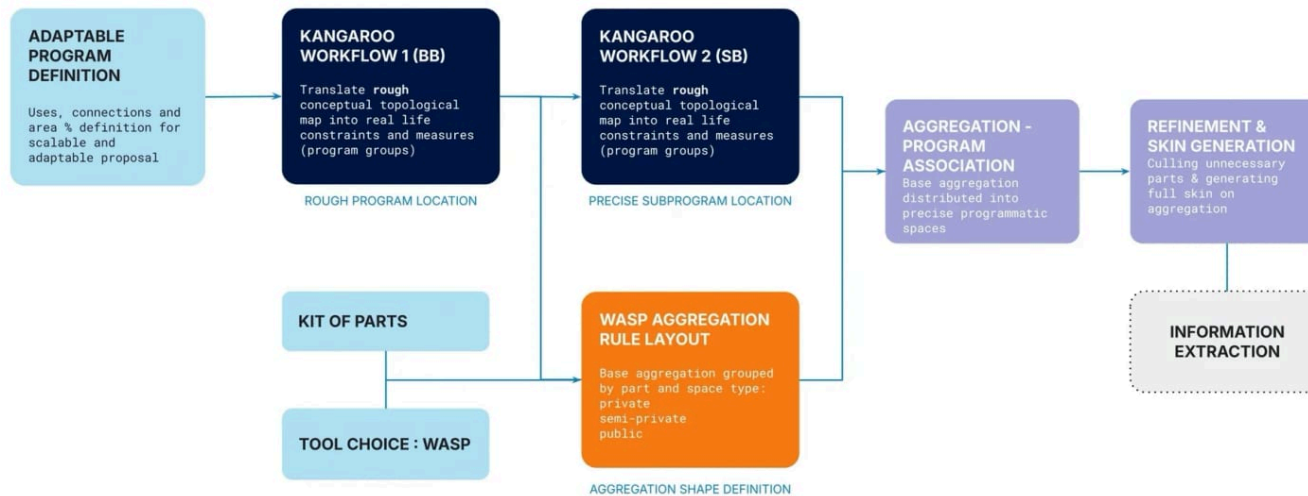
2025/26

DESIGN & RESEARCH

IAAC EXHIBITION

A research hub huddled against the wind.

Punta Arenas sits at the southern tip of Chile in a cold, perpetually windy subpolar climate. The Huddle is a research and education hub with an expedition basecamp for 1,000 users, designed the way penguins survive the same latitudes: compact forms, packed close, shielding one another from the wind.



MODULE

ACESD Studio · MacAD, IAAC

TEAM

Charles Abi Chahine, Emilie El Chidiac, María Sánchez Domínguez, Lakzhmy Mari Zaro

TOOLS

Rhino · Grasshopper · Wasp · Kangaroo · Speckle

## 01 · ENDURANCE, NOT SHELTER

Punta Arenas sits on the Magellan Strait, a departure point for Antarctic expeditions. We took the harder of two plots: open, windy, salt-worked, but facing the sea. The project accepts wind, not resists it.

## 02 · LEARNING FROM THE KAWI

The typology comes from the kawis, the region's native huts, already adapted to extreme wind. We reinterpreted it as 4x4x4 m units under a 16 m limit, the gaps becoming wind-protected courtyards.

## 03 · A WORKFLOW, NOT A SHAPE

One chain: a program definition feeds two Kangaroo solvers, then Wasp aggregates the kit of parts. Because it is a system, the part count is known at every moment and the rules hold at 50 users or 1,000.

## 04 · PROGRAM BECOMES MASSING

Public programme wraps around and shields the research and residential areas, on a single circulation spine. In section it stays compact under the height limit, the skin a buffer against the cold.

## 05 · TIMBER, SEISMIC FORCES, AND A SKIN OF THREE PANELS

Local timber, shear walls and bracing for wind, seismic isolators for movement, stable at roughly 709 tons. Three ETFE panels: the Shield for pressure, the Lens for solar gain, the Gill for ventilation.

## 06 · WIND BECOMES A RESOURCE

The tunnels the aggregation opens accelerate the flow, and vertical turbines stand in those zones. What the building was shaped to survive ends up powering it.

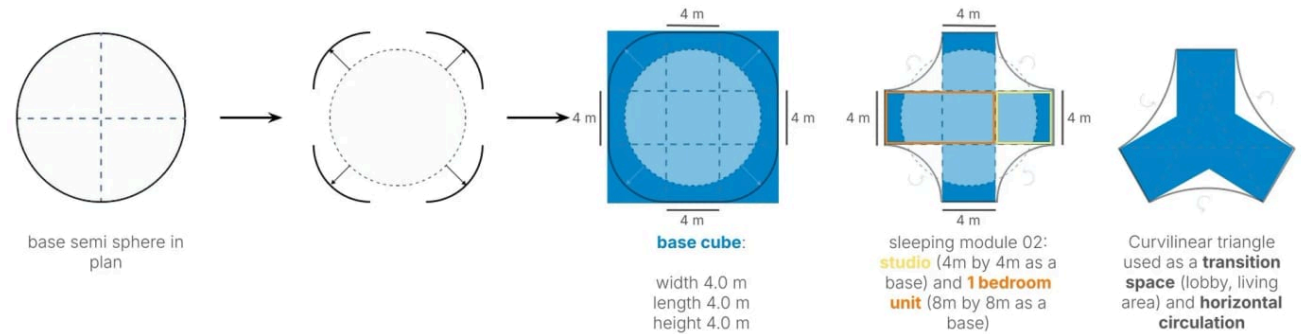


FIG 04.1

From semisphere to kit: the 4 m cube, the sleeping module, the triangle between them.

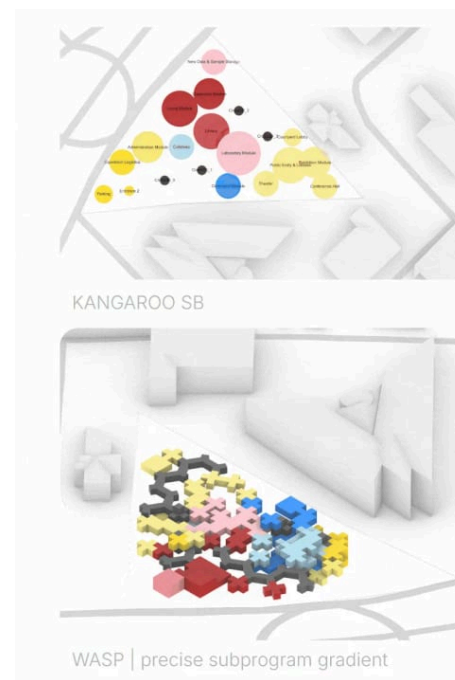


FIG 04.2

Mid-run: the Kangaroo gradient above, the Wasp aggregation below.

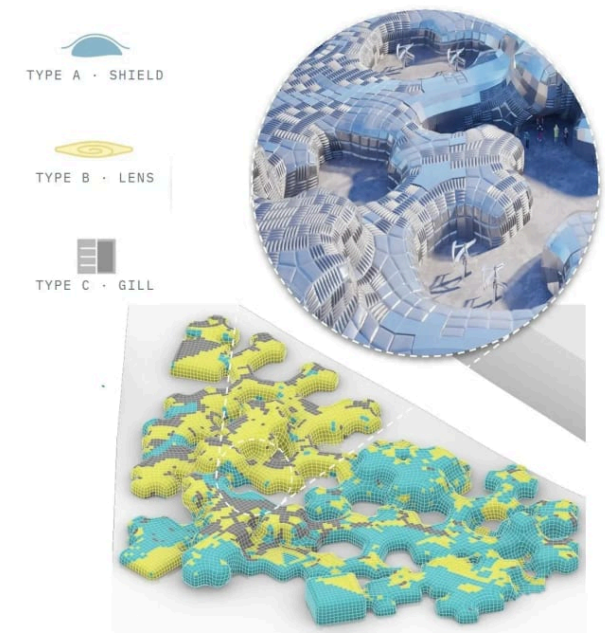


FIG 04.3

Panel placement as a morph: the skin in the render, and the mesh coloured by panel type from the daylight, orientation and wind indices.



# The Luminous Stratum

2025

DESIGN & RESEARCH

Sedimented light over a Cairo market.

In the dense historic fabric of Cairo, light is both a blessing and a burden. The Luminous Stratum proposes a new architectural language that negotiates this relationship: a “volume of sedimented light” that mimics the city’s stratification while filtering the harsh sun. Designed for the Complex Forming seminar, it is an independent system of stacked lamellas that hovers within the void of the Bab Al-Luk historic market, leaving the original structure untouched.



MODULE  
Complex Forming · MaCAD, IAAC

TEAM  
Charles Abi Chahine

TOOLS  
Rhino · Grasshopper · Kangaroo

**01 · MORE THAN A ROOF**

The system is deliberately independent: it revitalizes the market below without touching the historic walls. Voids are cut into the geometry precisely so the new form pulls away from its context: a porous buffer nested within the void.

**02 · COMPUTATIONAL FORM-FINDING**

The geometry is generated through Kangaroo physics. A flat mesh, strategic voids, edge anchor points and OnCurve goals define the footprint, then vertical load forces pull the mesh into a relaxed catenary vault. The defining move comes next: mesh faces are sorted by their normal vectors, and that orientation dictates function.

**03 · GEOMETRY BECOMES FUNCTION**

Vertical faces become gradient frosted louvers that control sun glare; horizontal faces become structural lamellas serving as shelving and walkable surfaces. The system is modular, and for the last vault different anchor points turn it into a continuous staircase.

**04 · AN OCCUPIABLE LATTICE**

I tested vertical loads from a shallow 15% to a steep 75%, and calibrated the domains for the frosted louvers and structural lamellas to tune porosity against structure. The selected iteration uses a 35% vertical load, giving the optimal vault height and clearance for the layers below.

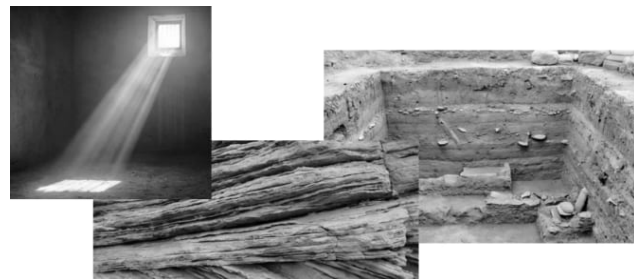


FIG 05.3

The references: a shaft of light in a dark room, sedimented strata, an excavated pit.

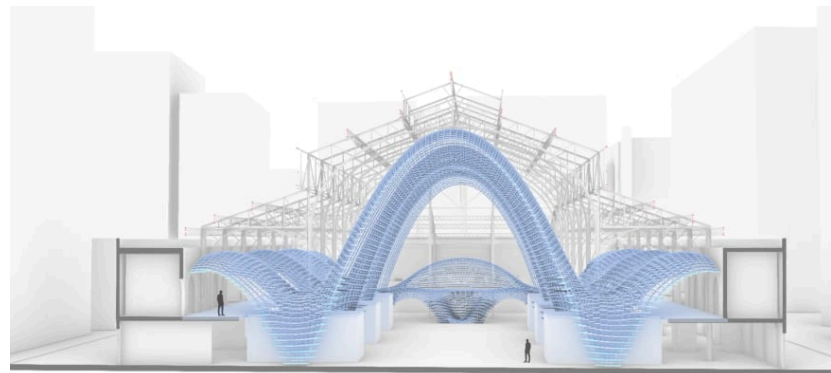


FIG 05.1

Section: an independent system hovering within the void, holding its distance from the historic walls.

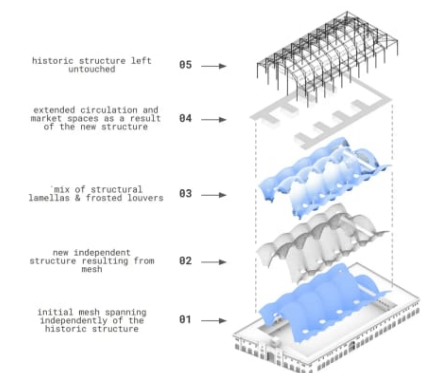


FIG 05.2

The system exploded, level by level: the historic hall untouched.

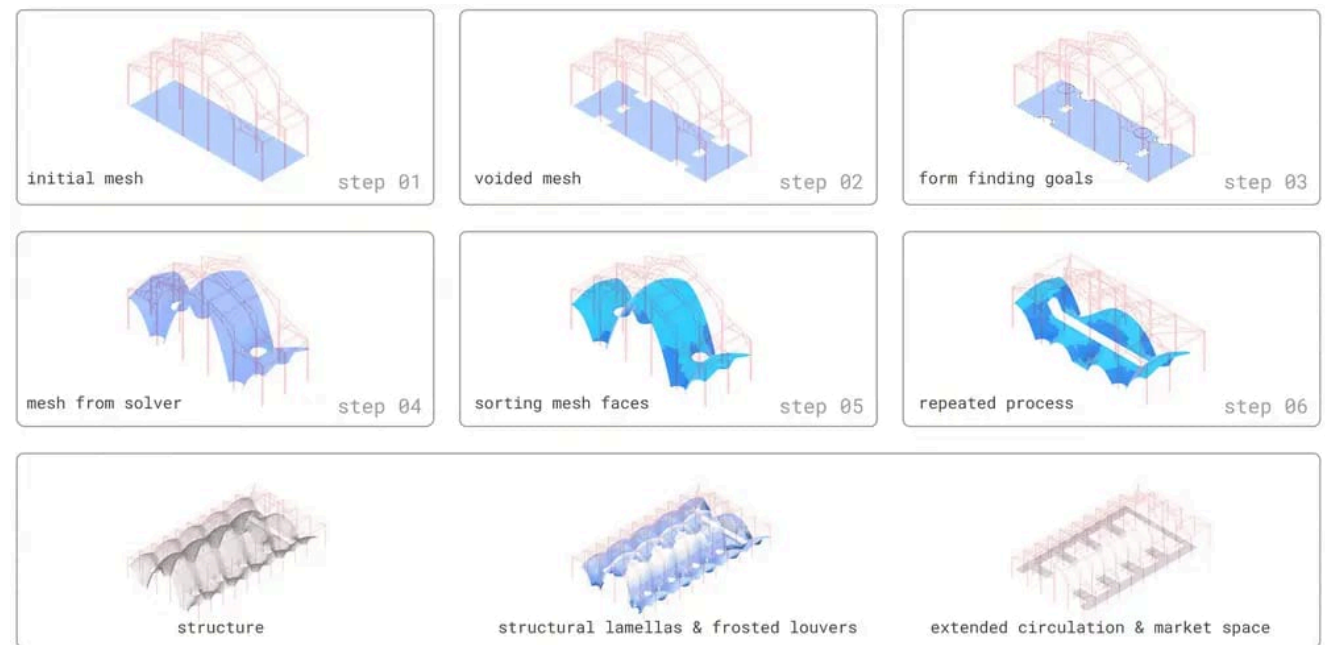


FIG 05.4 · DEMO

Step by step: flat mesh, strategic voids, form-finding goals, solver relaxation, face sorting.



## Rings of Mars: Ring 4000

2024

DESIGN & RESEARCH

TOP 50 · VOLUME ZERO

A regolith-printed habitat ring for Mars.

How do you design a habitat for Mars? Marsception, run by Volume Zero, asked that in 2024, and our answer was a ring. Ring 4000 sits on the rugged rim of a crater inside Valles Marineris, a 3D-printed regolith shell holding farming, research, living and a transportation ring in one closed loop, with the long game that every crater gets its own ring until the rings make a community. It was an early bet that generative tools belonged on an architect's desk, made when that still raised eyebrows.



MODULE  
International Competition

TEAM  
Charles Abi Chahine, Emilie El Chidiac

TOOLS  
Rhino · SubD · AI workflows · V-Ray

## 01 · A RING IN A CRATER

Three concentric circles: a farm under glass at the centre, visible from every room, transportation in the middle, living and working outside, five sleeping pods on the lowest level. The shell is printed regolith over an air-tight HDPE liner, with BIPV panels over the facade; the canyon walls do the rest of the shielding.

## 02 · WHY A RING

The craters are the site Mars already gives you, so the ring takes their shape. The ecosystem closes on itself: water split into oxygen and hydrogen, plants turning the crew's CO<sub>2</sub> into food, carbon and water reacting back into HDPE for printing and fuel. Transport runs in a loop, and expansion is the next crater over.

## 03 · HOW IT WAS MADE

Regolith melted into paste and printed by drones, every surface at an angle that needs no support. Modelled in Rhino with SubD, fed to generative image tools when that was still new, finished in V-Ray.

## 04 · WHAT CAME OF IT

Top 50 at Marsception, both our names on the results page, and our first proof that generative tools and architecture could share a desk.

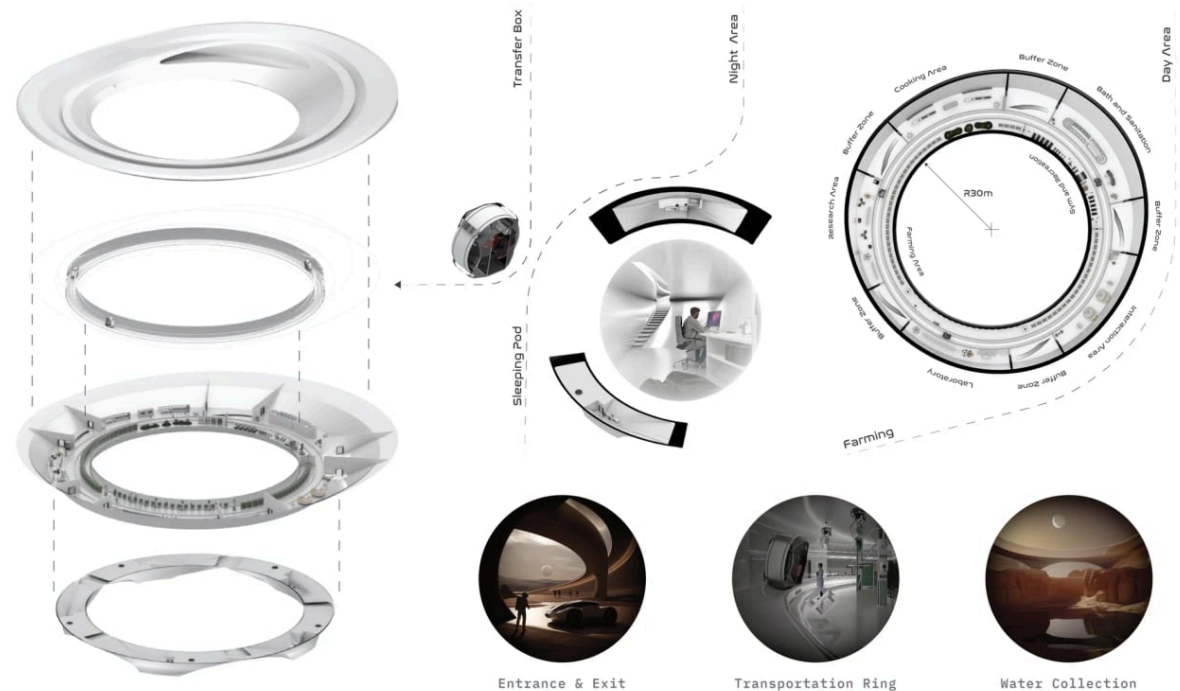


FIG 06.1

The ring by level, the shell to the base, with the transfer box, the sleeping pods and the plan of the outer circle at R30m.

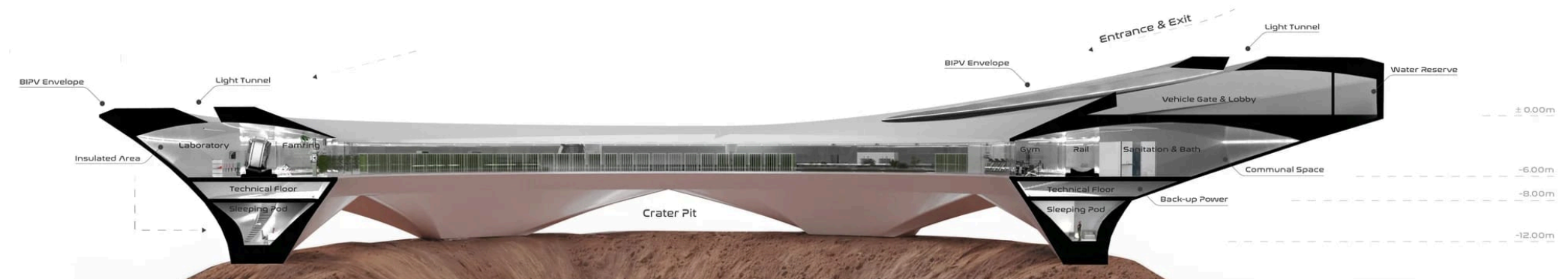


FIG 06.2

The long section through the crater pit: laboratory and farming on one side, gym, rail and the vehicle gate on the other, the sleeping pods and technical floors below.



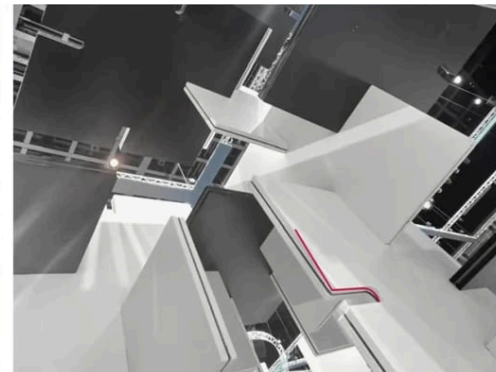
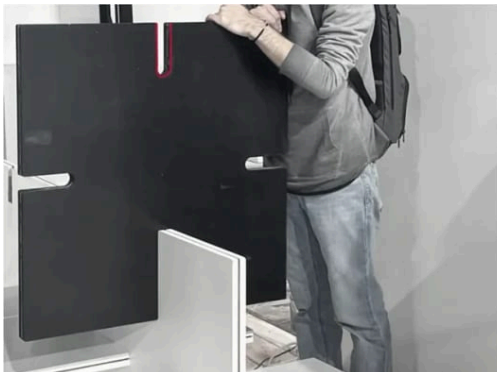
## Codependent

2024

PRACTICE

A booth grown from two puzzle pieces.

Codependent is an exhibition booth made from only two pieces: a white panel that always stands and a black one that always lies flat, each cut with slots that let it lock into the other. Rather than drawing the booth, we grew it. A form-finding tool in Grasshopper builds assemblies out of compatible joints inside a set of rules, lays the options out side by side, and turns the one we chose into a numbered kit of parts cut on a CNC.



### MODULE

Dynamic Solution Co. · Design  
Technology Architect

### TEAM

Dynamic Solution Co.: office project

### TOOLS

Rhino · Grasshopper · CNC

**01 · TWO PIECES**

Two panels,  $780 \times 780 \times 36$  mm: a white one that always stands, a black one that always lies flat. Slotted together with no glue or screws, they make the support, the seat and the counter.

**02 · GROWING THE BOOTH**

Grown from one white panel by extending, turning, rising or branching; a move that breaks the envelope, the clearance or the mass budget is retried.

**03 · MANY BOOTHS, ONE CHOSEN**

Thirty candidates grown, ten compared. Option 29 was chosen: 34 panels, a counter at 948 mm, a seat at 408 mm.

**04 · A KIT OF PARTS**

34 unique IDs, 23 white and 11 black, each with its position, mass, mating IDs and sheet; nested onto twelve sheets and cut on a CNC.

**05 · ON SITE**

Pieces went missing on site with no spares, so we reworked the connections with what we had.



FIG 07.1

Ten of the thirty candidates at one camera and scale; the chosen one outlined in red.

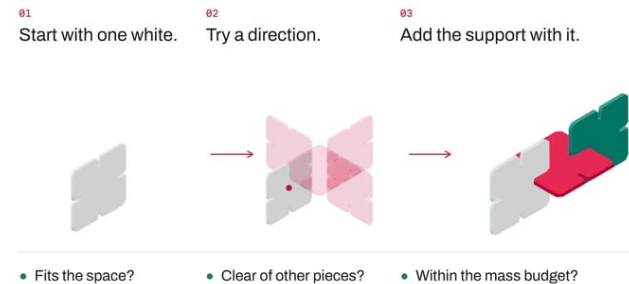


FIG 07.2

The growth rule: start with one white, try a direction, add the support with it, then check the space, the clearance and the mass.

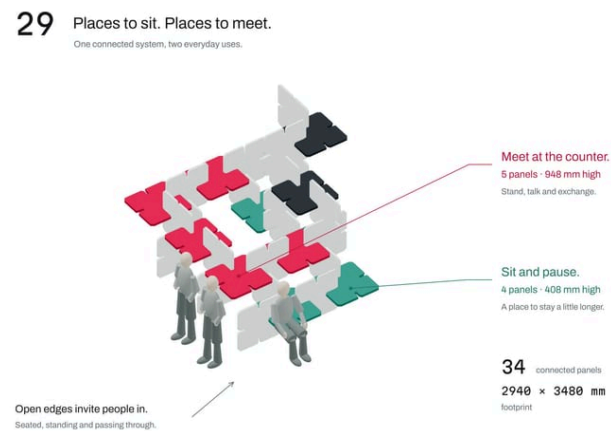


FIG 07.3

Option 29: counters at 948 mm, seats at 408 mm, 34 connected panels.

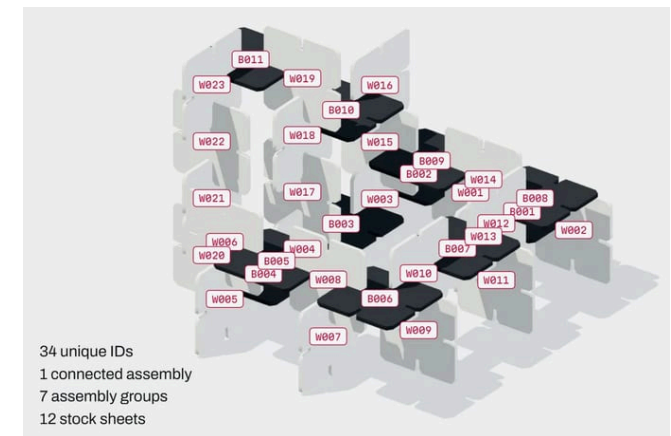


FIG 07.4

The kit: every panel labelled on the model by its ID, thirty-four parts nested onto twelve sheets.



[charlesabichahine.com/work/saria](https://charlesabichahine.com/work/saria)



## Saria, Dubai Maritime City

2024

PRACTICE

A residential tower on the water in Dubai.

Saria is a residential tower on the water at Dubai Maritime City, with the Dubai skyline behind it. I took it from pre-concept through to design development as a Design Architect at SOMA: parametric facade iterations in Rhino and Grasshopper, then the BIM model for delivery. The tower is now under construction. The drawings here are from the concept design submission of March 2024, with SOMA as design architect.

MODULE

SOMA · Design Architect

TEAM

SOMA: office project

TOOLS

Rhino · Grasshopper · Revit

## 01 · A TOWER ON THE WATER

The plot sits at the edge of the Maritime City peninsula, so the tower is read from the water first: a slender square plan rising off a podium. Four podium levels carry the parking under a landscaped deck, and 38 levels rise above them to a roof at 167 metres.

## 02 · THE FACADE

The facade is where the parametric work went. The balconies shift and step across the elevation rather than stacking in a line, iterated in Rhino and Grasshopper: how far each band projects, where the planters sit, how the rhythm breaks. What survived went into the Revit model.

## 03 · THE PODIUM AND THE GROUND

At street level the podium is wrapped in planting and opens under a deep canopy; the landscape runs over the podium roof down to the promenade, so the tower lands in a garden, not on a car park.

## 04 · THE DRAWINGS

The concept set: the master plan, the ground floor with the lobby and drop-off, the amenity level over the podium, and a typical floor from the 12th to the 34th, core in the middle.

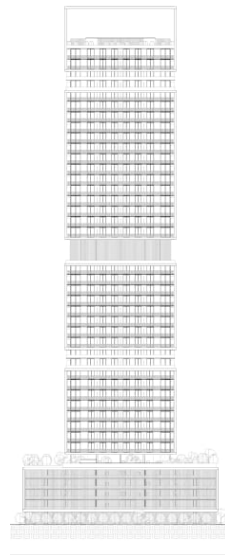


FIG 08.1

North-west elevation.

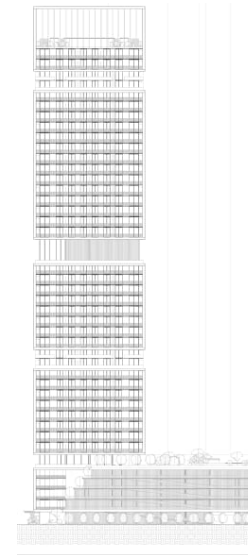


FIG 08.2

North-east elevation.

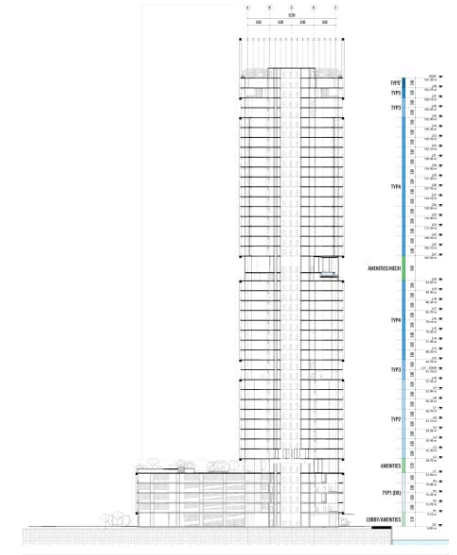


FIG 08.3

Section through the tower and podium.

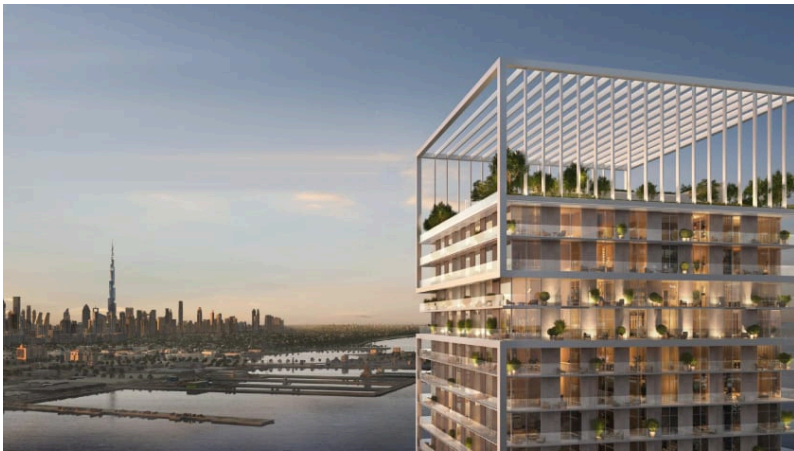


FIG 08.4

The crown against the Dubai skyline, the Burj Khalifa on the horizon.

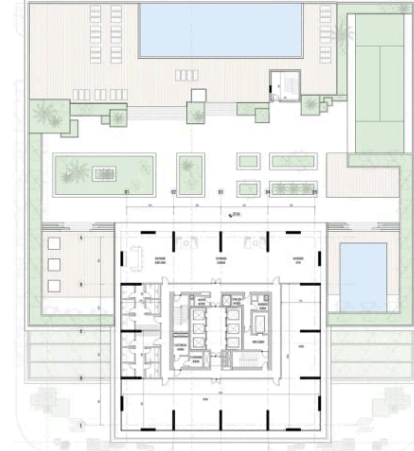


FIG 08.5

Level 01, the amenity floor over the podium: gym, outdoor deck and pool.

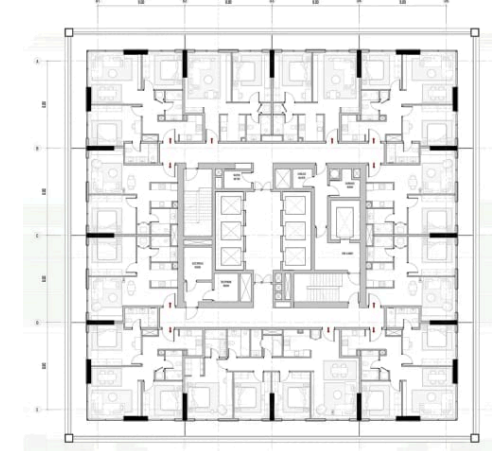


FIG 08.6

Typical floor: the core at the centre, apartments around it.

# The rest of it

This portfolio is a selection of eight from twenty. The other twelve, with every gallery, demo and write-up, are on the site.



## Narkomfin as a Graph

Centrality reveals the corridor as its spine.



## Breathing Mass

A tower that breathes for Santiago.



## Paper Trail

Model data shipped on every push.



## Revitalizing the Tsukiji Fish Market

The Tsukiji site put under Tokyo's sun.



## The Wave Chair

An undulating surface resolved into a chair.



## Tideline

A mirrored wave that hides a car, then lifts.



## FacadeIQ

Facade design priced as it is drawn.



## Family Tree

A skyscraper complex the script decides, down to the last panel.



## The Clebsch Pavilion

A pavilion solved on the Clebsch cubic.



## The Puffer Playscape

A playscape inflated from puffer geometry.



## The Cross-Cap House

A house on a self-intersecting surface.



## Point Nought

Anfeh reimagined through its salt ponds.





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